

# MATH 174 - HW 1

Dani Nguyen

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1. (a) Forward substitution on  $Ly = b$

$$\begin{aligned}x_1 &= 2 \\-x_1 + x_2 &= -1 \\3x_1 - 3x_2 + x_3 &= 3\end{aligned}$$

So, we have

$$\begin{aligned}x_1 &= 2 \\x_2 &= -1 + x_1 \\&= -1 + 2 \\&= 1 \\x_3 &= 3 - 3x_1 + 3x_2 \\&= 3 - 3(2) + 3(1) \\&= 0\end{aligned}$$

- (b) Backward substitution on  $Ux = y$

$$\begin{aligned}-1x_1 + 2x_2 + x_3 &= y_1 \\2x_2 - 2x_3 &= y_2 \\5x_3 &= y_3\end{aligned}$$

So, we have

$$\begin{aligned}
 x_3 &= \frac{y_3}{5} \\
 2x_2 &= y_2 + 2x_3 \\
 2x_2 &= y_2 + 2\left(\frac{y_3}{5}\right) \\
 x_2 &= \frac{1}{2}y_2 + \frac{y_3}{5} \\
 -x_1 &= y_1 - 2x_2 - x_3 \\
 x_1 &= -y_1 + 2y_2 + \frac{2y_3}{5} + \frac{y_3}{5} \\
 &= -y_1 + 2y_2 + \frac{3y_3}{5}
 \end{aligned}$$

(c) Compute  $A = LU$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ -1 & 1 & 0 \\ 3 & -3 & 1 \end{bmatrix} \begin{bmatrix} -1 & 2 & 3 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix} = \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix}$$

$$\begin{aligned}
 Ax = b : \quad & \begin{bmatrix} -1 & 2 & 3 \\ 1 & 0 & -5 \\ -3 & 0 & 20 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix} \\
 & \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}
 \end{aligned}$$

The solution to  $Ax + b$  is any multiple of  $\begin{bmatrix} -1 \\ \frac{1}{2} \\ 0 \end{bmatrix}$

2. (a) The summation is from  $j = 1$  to  $i - 1$ , there are  $i - 1$  additions. There is also one more subtraction between  $b_i$  and the summation. So in total, there are  $i$  additions and subtractions.
- (b) There are  $i - 1$  terms in the summation, therefore there are  $i - 1$  multiplications.

3. (a) Gaussian elimination

$$\begin{aligned}
 & \left[ \begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ -4 & 0 & 4 & -1 \end{array} \right] \\
 & \left[ \begin{array}{ccc|c} 2 & -3 & 1 & 1 \\ 1 & 1 & -1 & 2 \\ 0 & -6 & 6 & 1 \end{array} \right] R_3 = R_3 + 2R_1 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 1 & 1 & -1 & 2 \\ 0 & -6 & 6 & 1 \end{array} \right] R_1 = R_1 - R_2 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 5 & -3 & 3 \\ 0 & -6 & 6 & 1 \end{array} \right] R_2 = R_2 - R_1 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & -1 & 3 & 4 \\ 0 & -6 & 6 & 1 \end{array} \right] R_2 = R_2 + R_3 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & -6 & 6 & 1 \end{array} \right] R_2 = -R_2 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & -12 & -23 \end{array} \right] R_3 = R_3 + 6R_2 \\
 & \left[ \begin{array}{ccc|c} 1 & -4 & 2 & -1 \\ 0 & 1 & -3 & -4 \\ 0 & 0 & 1 & \frac{23}{12} \end{array} \right] R_3 = \frac{-1}{12}R_3
 \end{aligned}$$

Back substitution

$$x_3 = \frac{23}{12}$$

$$x_2 = 3x_3 - 4$$

$$= \frac{23}{4} - 4$$

$$= \frac{7}{4}$$

$$x_1 = 4x_2 - 2x_3 - 1$$

$$= 7 - \frac{23}{6} - 1$$

$$= \frac{13}{6}$$

4. (a) Gaussian elimination

$$\left[ \begin{array}{cc|c} 1 & 4 & 3 \\ 0.002 & -4 & -2 \end{array} \right]$$

$$\left[ \begin{array}{cc|c} 1 & 4 & 3 \\ 0 & 4 & -1 \times 10^3 \end{array} \right]$$

(b)

(c)

5.

6.

7.

8.